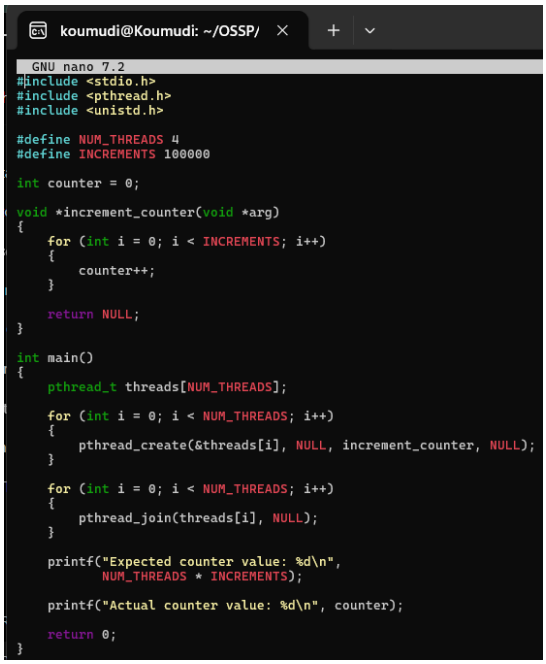
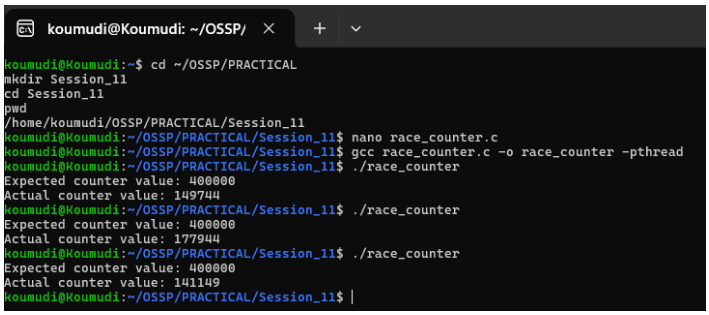
Practical Week 11

Session 11:

Task1: Develop a multithreaded counter application using POSIX threads (pthread create, pthread_ioin). Demonstrate race conditions by allowing multiple threads to update a shared variable concurrently.





Task2: Modify the previous program using mutex locks (pthread mutex lock, pthread mutex_unlock) and compare the outputs with the race- condition version. Analyze the impact of synchronization.

koumudi@Koumudi: ~/OSSP/

GNU nano 7.2

```c
#include <stdio.h>
#include <pthread.h>

#define NUM_THREADS 4
#define INCREMENTS 100000

int counter = 0;

pthread_mutex_t lock;

void *increment_counter(void *arg)
{
    for (int i = 0; i < INCREMENTS; i++)
    {
        pthread_mutex_lock(&lock);

        counter++;

        pthread_mutex_unlock(&lock);
    }

    return NULL;
}

int main()
{
    pthread_t threads[NUM_THREADS];

    pthread_mutex_init(&lock, NULL);

    for (int i = 0; i < NUM_THREADS; i++)
    {
        pthread_create(&threads[i], NULL, increment_counter, NULL);
    }

    for (int i = 0; i < NUM_THREADS; i++)
    {
        pthread_join(threads[i], NULL);
    }

    printf("Expected counter value: %d\n",
           NUM_THREADS * INCREMENTS);

    printf("Actual counter value with mutex: %d\n", counter);

    pthread_mutex_destroy(&lock);

    return 0;
}
```

```
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_11$ nano mutex_counter.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_11$ gcc mutex_counter.c -o mutex_counter -pthread
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_11$ ./mutex_counter
Expected counter value: 400000
Actual counter value with mutex: 400000
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_11$ ./mutex_counter
Expected counter value: 400000
Actual counter value with mutex: 400000
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_11$ ./mutex_counter
Expected counter value: 400000
Actual counter value with mutex: 400000
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_11$ echo "----- Race Condition Version -----"
./race_counter

echo "----- Mutex Version -----"
./mutex_counter
----- Race Condition Version -----
Expected counter value: 400000
Actual counter value: 205660
----- Mutex Version -----
Expected counter value: 400000
Actual counter value with mutex: 400000
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_11$ ls -lh
total 44K
-rwxr-xr-x 1 koumudi koumudi 17K Sep 24 16:37 mutex_counter
-rw-r--r-- 1 koumudi koumudi 837 Sep 24 16:37 mutex_counter.c
-rwxr-xr-x 1 koumudi koumudi 16K Sep 24 16:34 race_counter
-rw-r--r-- 1 koumudi koumudi 676 Sep 24 16:34 race_counter.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_11$
```